E-commerce Sales Data Analysis

Project Report

Products, months, regions, and sales and profit trends

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Tools: Python, Pandas, NumPy, SQL (SQLite), Matplotlib, Seaborn, Jupyter Notebook

1. Objective

This project analyses the sales performance of an online store to support business decisions. It answers four questions:

- Which products generate the highest revenue?
- Which months have the highest sales?
- Which regions perform best?
- What are the sales and profit trends?

2. Data

The analysis uses the Sample Superstore dataset (Kaggle): 9,994 order lines and 21 columns, covering 03 January 2014 to 30 December 2017. Key columns are Order Date, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount and Profit.

Data quality: the dataset has 0 missing values and 0 duplicate rows. The same Order ID appears on several rows because each product in an order has its own row; this is expected and not a duplicate. The only cleaning step was converting Order Date to a datetime type.

3. Methodology

- **Pandas and NumPy:** grouped and aggregated sales and profit by product, month, region, year and category.
- **SQL (SQLite):** loaded the data into a SQLite database and wrote equivalent queries for each question (saved in queries.sql). SQL results were compared with the Pandas results to confirm they match.
- **Visualization:** Matplotlib and Seaborn charts, saved as PNG files.
- **Deeper analysis:** discount bands against profit, and profit by sub-category and product.

4. Findings

Overall performance

Across the four years the store made **$2,297,200.86** in sales and **$286,397.02** in profit, an overall profit margin of **12.47%**.

4.1 Which products generate the highest revenue?

The highest-earning product is the **Canon imageCLASS 2200 Advanced Copier** with $61,599.82 in sales, more than double the second-placed product (*Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind*, $27,453.38). Office machines and equipment such as copiers, binding machines, videoconferencing units and printers dominate the top 10.

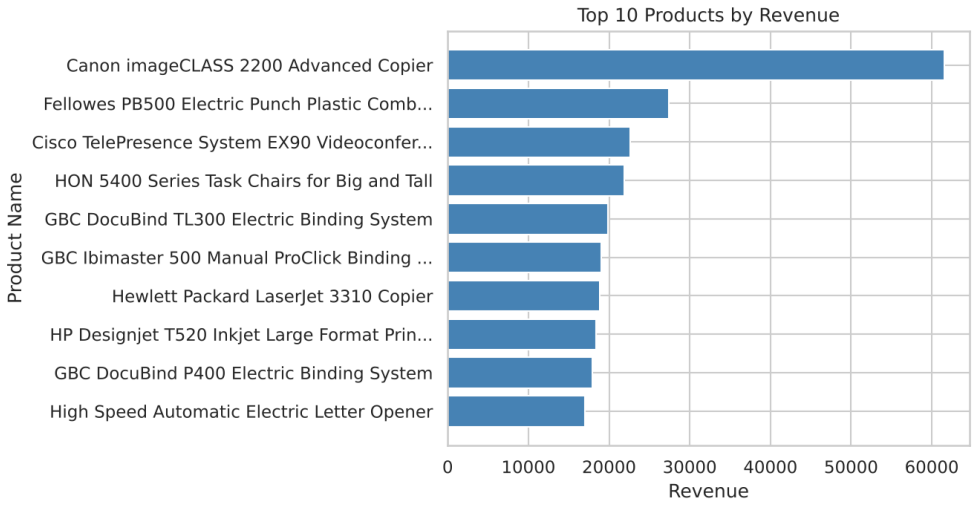


Figure 1: Top 10 products by revenue

4.2 Which months have the highest sales?

November ($352,461.07), December ($325,293.50) and September ($307,649.95) are the strongest months. These three months rank in the top three in every year from 2014 to 2017, so the pattern is consistent and not driven by one strong year. January ($94,924.84) and February ($59,751.25) are the weakest, which suggests holiday-season demand followed by a post-holiday slump.

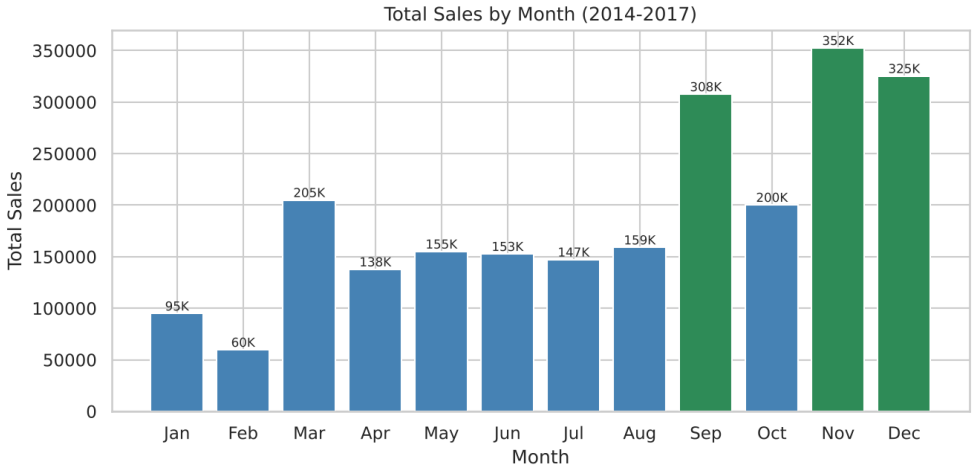


Figure 2: Total sales by month, top three months highlighted

4.3 Which regions perform best?

The **West** is the best performer on sales, profit and margin. The **Central** region has higher sales than the South but the lowest profit and by far the lowest margin, which points to heavy discounting or low-margin products in that region.

Region	Sales	Profit	Profit margin
West	$725,457.82	$108,418.45	14.94%
East	$678,781.24	$91,522.78	13.48%
Central	$501,239.89	$39,706.36	7.92%
South	$391,721.91	$46,749.43	11.93%

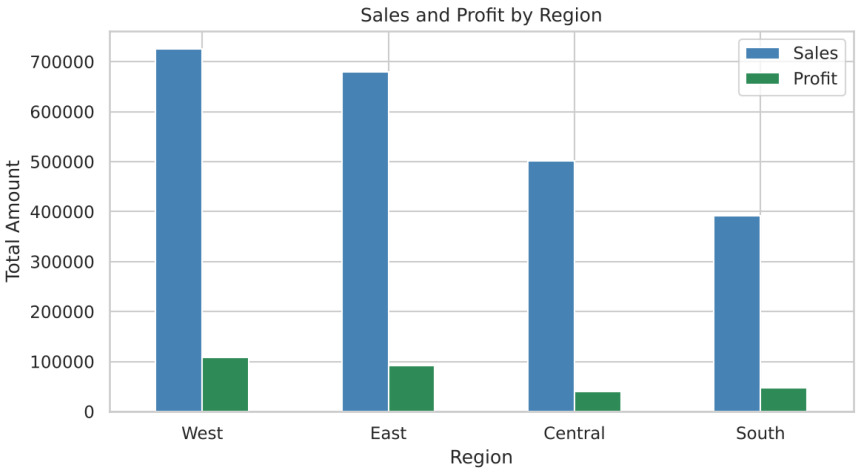


Figure 3: Sales and profit by region

4.4 What are the sales and profit trends?

Sales dipped slightly in 2015, then grew every year, rising about 51% from 2014 to 2017. Profit grew faster, about 89%, because the margin improved from 10.2% in 2014 to roughly 13% afterwards. Monthly profit is volatile; the largest monthly loss was in January 2015 (-$3,281.01).

Year	Sales	Profit	Margin
2014	$484,247.50	$49,543.97	10.2%
2015	$470,532.51	$61,618.60	13.1%
2016	$609,205.60	$81,795.17	13.4%
2017	$733,215.26	$93,439.27	12.7%

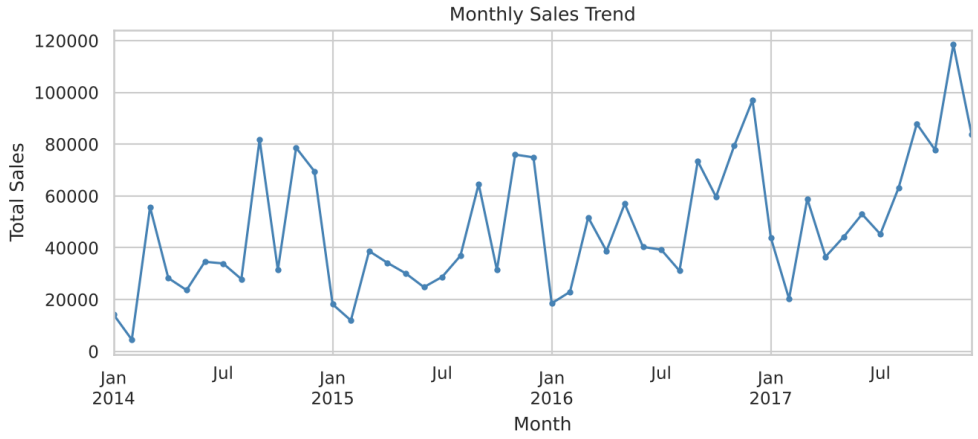


Figure 4: Monthly sales trend

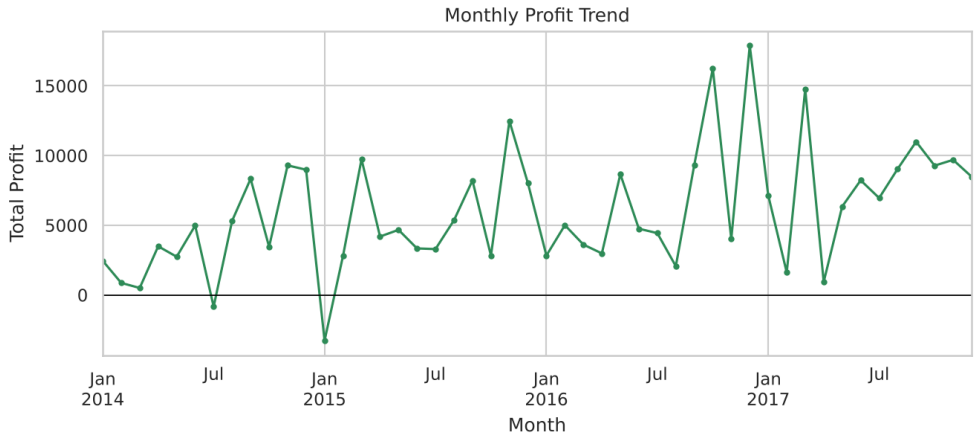


Figure 5: Monthly profit trend

5. Deeper Insights

Discounts and profit

There is a mild negative correlation (-0.22) between discount and profit. Grouping order lines by discount level shows where it turns harmful: **discounts above 20% lose money on average** (average profit per order line: $66.90 at 0%, $26.50 at 1-20%, -$77.86 at 21-40%, and -$106.71 above 40%).

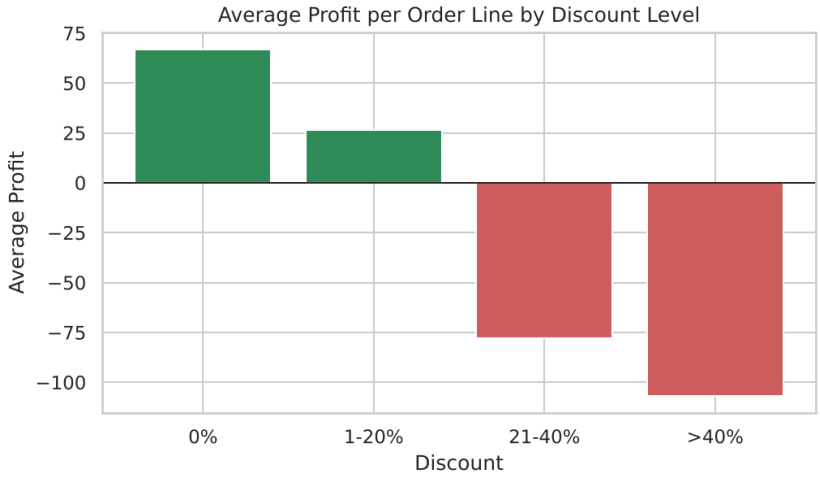


Figure 6: Average profit by discount level

Profit by sub-category and product

Three sub-categories lose money overall: **Tables (-$17,725.48)**, **Bookcases (-$3,472.56)** and **Supplies (-$1,189.10)**. The most profitable are Copiers ($55,617.82), Phones ($44,515.73) and Accessories ($41,936.64). Technology ($145,454.95) and Office Supplies ($122,490.80) earn most of the profit, while Furniture makes only $18,451.27 on $741,999.80 in sales, a margin of about 2.5%. The biggest single-product losses are the Cubify CubeX 3D Printer Double Head Print (-$8,879.97) and the Lexmark MX611dhe Monochrome Laser Printer (-$4,589.97).

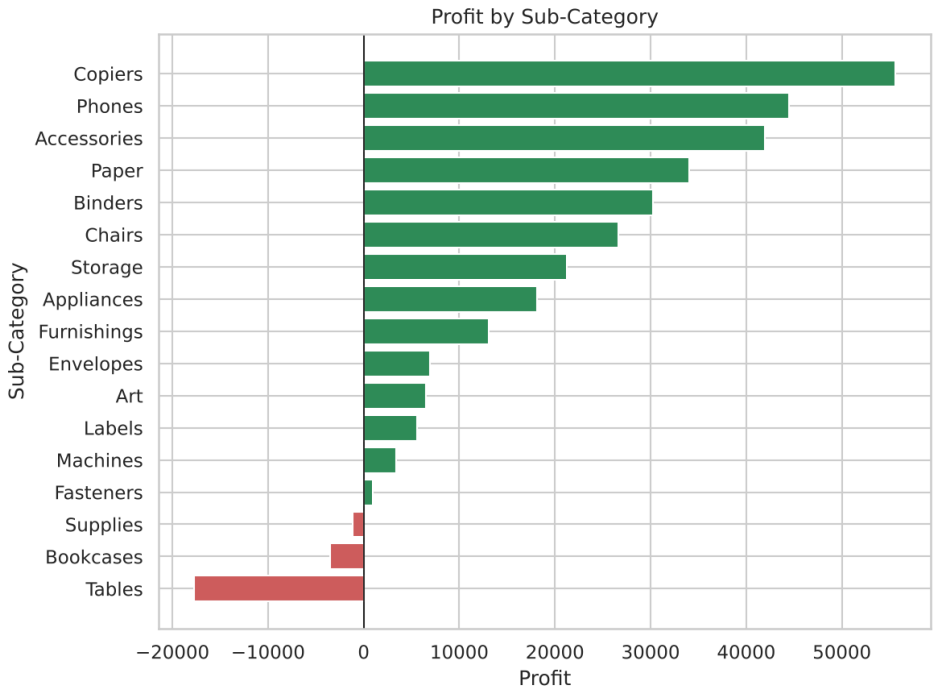


Figure 7: Profit by sub-category

6. Recommendations

- **Cap discounts at around 20%.** Deeper discounts lead to losses on average.
- **Review pricing on Tables and Bookcases,** which lose money despite high sales.
- **Plan for September to December.** Build inventory and run campaigns ahead of the peak months, and consider promotions in January and February to lift the slow period.
- **Investigate the Central region.** Its margin is far below the West and East, so review its discounting and product mix.
- **Keep promoting high-revenue equipment lines** (copiers, binding machines, printers) that drive a large share of sales.

7. Limitations

- The analysis uses a single dataset with no customer-level behaviour or marketing data.
- Monthly totals combine four years, so one unusually strong year can influence the ranking (although the top months are consistent across years).
- Sales and profit are in the dataset's original currency, with no inflation adjustment.
- A correlation between discount and profit does not prove that discounts cause lower profit.